

Hypo-Stage: Tool-Supported Architectural Hypothesis Engineering in Professional Software Development Teams — A Case Study

José Gonçalves Lima Neto

May 2026

Master's Thesis Defense

Graduate Program in Computer Science

IME – University of São Paulo





Overview

- ➊ Introduction
- ➋ Background
- ➌ Related Work
- ➍ Research Design
- ➎ Results
- ➏ Conclusion and Future Work
- ➐ References



Overview

- 1 Introduction
- 2 Background
- 3 Related Work
- 4 Research Design
- 5 Results
- 6 Conclusion and Future Work
- 7 References

Motivation

- Organizations strive for fast flow through autonomous teams
- Architecture decisions often remain centralized in a few experts
- Architectural uncertainty is natural but usually **implicit**
- Existing derivation methods rely on architects' tacit knowledge (Souza et al., 2019)
- Teams without senior architects lack guidance and tooling

The Problem

Gap

No consolidated approach exists for managing architectural uncertainty within the daily engineering workflow of development teams.

- ArchHypo (Silva et al., TSE 2025) uses hypotheses engineering to make uncertainty explicit and manageable
- But: manual application has a **steep learning curve** (Finding 9: unanimously hard to learn)
- Process-change resistance (Finding 8) hinders adoption

Objective

Thesis Objective

Empirically evaluate and refine **Hypo-Stage**, a support tool for the ArchHypo technique, by applying it with diverse software engineering teams in a real-world organization with a scaled software product.

Overview

- 1 Introduction
- 2 Background**
- 3 Related Work
- 4 Research Design
- 5 Results
- 6 Conclusion and Future Work
- 7 References

ArchHypo Technique

- ① Identify architectural hypotheses (uncertainties)**
 - ② Assess each: uncertainty level + impact level**
 - ③ Plan technical actions (spikes, tracer bullets, triggers)**
 - ④ Execute plans and reassess iteratively**
-
- Prior evaluations: Catch Solve startup (IEEE SW), PropSEP at Jetsoft (TSE)**
 - Both manual — no tool support**

Hypo-Stage Tool

- **Open-source Backstage plugin for ArchHypo**
- **Hypothesis CRUD, Likert-scale assessment, technical plans**
- **Visualization: uncertainty vs. impact chart**
- **Integrates into the Internal Developer Portal workflow**

Lineage

Initial design & implementation: undergrad capstone (Corrêa, 2025).
This thesis: empirical evaluation & refinement with real teams.

Overview

- 1 Introduction
- 2 Background
- 3 Related Work**
- 4 Research Design
- 5 Results
- 6 Conclusion and Future Work
- 7 References

Related Work

- ADR tools (adr-tools, Log4brains) — capture decisions, not uncertainties
- ATAM workshops — heavyweight, not continuous
- IDPs (Backstage, Port, Cortex) — no native hypothesis support
- Gap: no existing tool combines hypothesis engineering with developer portal integration

Overview

- 1 Introduction
- 2 Background
- 3 Related Work
- 4 Research Design**
- 5 Results
- 6 Conclusion and Future Work
- 7 References

Empirical Approach

- **Context:** CompanyX, scaled product, multiple teams
- **Method:** embedded case study, mixed methods
- **Data sources:**
 - Hypo-Stage artifacts (hypotheses, assessments, plans)
 - Questionnaires (usability, perceived value)
 - Semi-structured interviews
- **Analysis:** thematic coding + descriptive statistics

Overview

- 1 Introduction
- 2 Background
- 3 Related Work
- 4 Research Design
- 5 Results**
- 6 Conclusion and Future Work
- 7 References

(to be presented after the empirical study)

Overview

- 1 Introduction
- 2 Background
- 3 Related Work
- 4 Research Design
- 5 Results
- 6 Conclusion and Future Work**
- 7 References

Conclusion and Future Work

- **Summary of contributions**
- **Limitations**
- **Future directions**

Overview

- 1 Introduction
- 2 Background
- 3 Related Work
- 4 Research Design
- 5 Results
- 6 Conclusion and Future Work
- 7 References**

References i

- ▶ Bass, Len et al. (2021). *Software Architecture in Practice*. 4th ed. Addison-Wesley.
- ▶ Corrêa, Pedro Henrique Mariano (2025). *ArchHypo Tooling: Enabling Practical Hypothesis Engineering for Software Architecture*. Capstone Project Report (Bachelor), Institute of Mathematics and Statistics, University of São Paulo. Supervised by José G. Lima Neto; co-supervised by Paulo Meirelles.
- ▶ Foote, Brian and Joseph Yoder (1999). “Big Ball of Mud”. *Pattern Languages of Program Design* 4, pp. 654–692.
- ▶ Richards, Mark and Neal Ford (2020). *Fundamentals of Software Architecture: An Engineering Approach*. O’Reilly Media. ISBN: 9781492043454.
- ▶ Silva, Eduardo, Jorge Melegati, Moises Silveira, et al. (2025). “ArchHypo: Managing Software Architecture Uncertainty Using Hypotheses Engineering”. *IEEE Transactions on Software Engineering* 51.2, pp. 481–499. DOI: 10.1109/TSE.2024.3520477.

- ▶ Silva, Eduardo, Jorge Melegati, Xiaofeng Wang, et al. (2024). “Using Hypotheses to Manage Technical Uncertainty and Architecture Evolution in a Software Start-up”. *IEEE Software* 41.4, pp. 78–87. DOI: 10.1109/MS.2024.3383628.
- ▶ Skelton, Matthew and Manuel Pais (2019). *Team Topologies: Organizing Business and Technology Teams for Fast Flow*. IT Revolution Press.
- ▶ Souza, Eric et al. (2019). “Deriving Architectural Models from Requirements Specifications: A Systematic Mapping Study”. *Information and Software Technology* 109, pp. 26–39. DOI: 10.1016/j.infsof.2019.01.004.

Hypo-Stage

- ➊ Introduction
- ➋ Background
- ➌ Related Work
- ➍ Research Design
- ➎ Results
- ➏ Conclusion and Future Work
- ➐ References





Backup: ArchHypo Nine Findings (TSE)

- ❶ Some hypotheses monitored without full resolution until end
- ❷ Avoided upfront architectural design; decisions through iterations
- ❸ Improved process: safety, transparency, predictability
- ❹ Different QAs handled with different creative strategies
- ❺ Guideline implementation was the most-used strategy
- ❻ Team recognized better architectural decision-making
- ❼ Evidence of positive impact on product quality
- ❽ **Process-change obstacle:** mixing arch & functional tasks
- ❾ **Unanimously hard to learn:** mapping, articulating, planning

Backup: Souza et al. Key Findings

- **2575 papers → 39 primary studies**
- **All methods rely on architects' tacit knowledge**
- **Limited support for inexperienced practitioners**
- **Weak or absent tool support**
- **Insufficient empirical evaluation**